OpenTPS: an open-source treatment planning system to foster research and innovation.

A. Barragán-Montero[1], D. Dasnoy[2], S. Wuyckens[1], W. Zhao[2], E. Peeters[1], R. Schyns[1], M. Huet-Dastarac1, E. Löyen[2], G. Rotsart de Hertaing[2], G. Janssens[3], V. Hamaide[3], S. Deffet[3], K. Souris[3], E. Sterpin[1,4], B. Macq[2], J. A. Lee [1,2]

[1]Molecular Imaging Radiation Oncology (MIRO), UCLouvain, Brussels, Belgium
[2] ICTeam, UCLouvain, Louvain-La-Neuve, Belgium
[3] Ion Beam Applications S.A, Louvain-La-Neuve, Belgium
[4] Department of Radiation Oncology, KULeuven, Leuven, Belgium

Abstract Treatment planning systems (TPS) are an essential component for simulating and optimizing a radiation therapy treatment before administering it to the patient. However, TPS commercialized by private companies are often black-boxes when it comes to the implementation of most of their features (i.e. image registration, dose calculation and optimization algorithms). This hampers and delay researchers when working with these TPS for implementing and testing new ideas. We have developed OpenTPS, a new open-source software platform (opentps.org), to foster high quality research involving treatment planning for external beam radiotherapy, and, in particular, for proton therapy. It is designed to be a flexible and user-friendly platform for medical physicists, radiation oncologists, and other members of the radiation therapy and research community. It serves to create customized treatment plans for educational and research purposes, particularly for emerging delivery and optimization techniques such as FLASH, arc proton therapy, or beamlet free optimization, among others.

1 Introduction

Planning a radiotherapy treatment is a complex task, which requires the acquisition of advanced medical images and the use of sophisticated algorithms to analyze these images, simulate the effect of radiation on the patient (i.e. dose calculation) and optimize the radiation dose distribution. Today, the main treatment planning systems (TPS) are marketed by big companies, which charge up to several hundred thousand euros for a license of such software. They come with multiple features and a user-friendly GUI, but their big drawback is their low modularity, lack of transparency accessibility to the code. This difficulties their use in research environments, sometimes delaying or hampering research projects. This is the case, for example, of projects that attempt to study treatment planning for new dose delivery modalities, faster optimization algorithms, more precise treatments, etc. Additionally, small research groups and clinics that do not have access to these expensive TPS experience a greater difficulty in conducting relevant research in the field, further slowing scientific progress in the field.

To address this issue, multiple groups have contributed to the development of open-source tools for treatment planning over the past decades. Well-known platforms are matRad [1], FoCa [2], CERR [3], or some modules embedded in 3DSlicer [4]. Our group has also a long experience in open-source tools, with the development of REGGUI [5] for image-processing, MCSquare [6] for Monte Carlo dose calculation in proton therapy, and MIROpt [7] for robust optimization in proton therapy. While these tools were entirely developed in Matlab and/or C++, in 2020 we embarked on a project to merge these existing tools and develop our own open-source platform, OpenTPS, using the state-of-the-art and widely-available Python programming language. OpenTPS (opentps.org) is designed to serve as a flexible framework for both personal and collective research efforts. Its open-source code is publicly available under the Apache-2.0 license. The code is designed to be easily extended and customized. The software is organized into two packages: the Core package, which is a library that defines data classes, data processing methods, and IO methods, and the GUI package, which offers a graphical interface for viewing and interacting with the data. In the current release, the Core package includes a range of features that are essential for proton therapy treatment planning, but extension to conventional radiotherapy treatments is being addressed and will be included in the next release. Some of the key features available in the current Core package of OpenTPS include:

- Data management and processing: DICOM importing, exporting and processing patient data.
- Dose computation: computing the dose from proton therapy treatment plans using fast Monte Carlo simulations with MCsquare.
- Treatment planning: creating treatment plans for individual patients, including the ability to define beam angles, energy levels, spot spacing, etc.
- Treatment evaluation: performing quality assurance checks on treatment plans, including the ability to evaluate robustness, compare doses, and assess the conformity of the treatment plan to the patient's anatomy.
- Data augmentation: creating artificial variations of patient data for robustness evaluation and for machine learning applications.

In the next section, we detail these tools and give an example of their use for two use cases: 1) implementation and evaluation of advanced optimization techniques of arc proton therapy, and 2) generation of synthetic data to simulate and evaluate different scenarios of tumor motion.

2 Materials and Methods

2. 1. GUI

OpenTPS comes with an optional graphical user interface (GUI) for visualization of 3D images, dose maps, contours and dynamic images. Several other interactive tools are available: profile plots, visualization of DVH, segmentation, etc. The implementation is based on the PyQt, VTK, and PyQtGraph libraries, which are widely recognized for their capabilities in data visualization and analysis in scientific and medical applications. Figure 1 shows a screenshot of the GUI loaded with a head-and-neck patient geometry and the optimized treatment dose associated to the case. In addition, graphical modules have been implemented to design and optimize a treatment plan. The entire GUI is extensible and modifiable by the user. Examples of extensions can be found in the OpenTPS documentation.

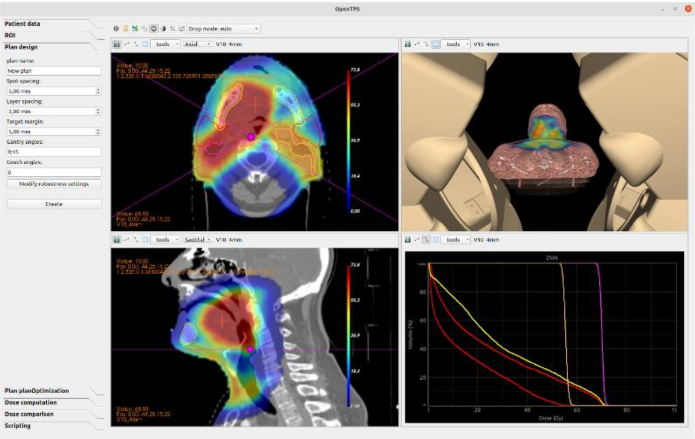

Figure 1. GUI with a head-and-neck cancer patient and an intensity-modulated-proton-therapy dose distribution.

2. 2. Core package

Data management and processing

The current version of OpenTPS supports the import patient data in DICOM format (CT, RT structures, RT dose, RT ion plan and vector field), MHD format (3D images), and OpenTPS-specific serialized object format. DICOM and MHD export is also supported.

The core library of OpenTPS includes a range of common image processing methods: resampling, coordinate sys- tem conversion, affine transformations, and filtering. Additionally, it has segmentation options such as automatic segmentation for body, lung and bones in CT images.

Rigid and deformable image registration methods are available, including the diffeomorphic Morphons algorithm [8]. The morphons registration is also used to create motion models. Deformation fields between the 4DCT phases are computed. Then with the deformations from all 4DCT phases, the midposition (MidP) image [9] can be computed. A motion model, called Dynamic3DModels in OpenTPS, is composed of the MidP image and the various deformation fields from the MidP image to each of the 4DCT phases.

Digitaly reconstructed radiographs (DRR) can also be computed using a 3DCT to simulate in room x-ray devices.

Dose calculation

A stable release of the fast Monte Carlo for proton therapy, MCsquare, is included in OpenTPS. However, OpenTPS is designed with the flexibility to support multiple dose engines and we are currently working on the implementation of photon dose calculation algorithms, both Monte Carlo and analytical.

MCSquare is available via the class MCsquareDoseCalculator and can be used to compute dose distributions, dose influence matrices, and LET distributions. Input paramaters such as the number of primaries, the beam model and the CT calibration are required to compute a dose based on CT and a treatment plan. From those inputs, MCsquareDoseCalculator generates the files required by MCsquare. Then, MCsquare simulates the interaction of each particle in batches until it reaches a user-defined minimal statistical uncertainty, or it simulates all the primaries.

Plan optimization

The treatment planning module of OpenTPS provides a set of tools and functions that are used to create and optimize a treatment plan based on several inputs. In general, the method followed depends on the treatment delivery modality chosen. As OpenTPS is initially built for research in proton therapy, the implementation of the treatment planning module conform to IMPT delivery, but implementation of IMRT and VMAT is being investigated. In proton pencil beam scanning (PBS), the dose is painted using scanning magnets that steer each pencil beam to a specific location in the target volume, namely a spot. Dividing the tumor into iso-energy layers, the treatment is therefore delivered by irradiating the volume spot-by-spot and layer-by-layer. This process is repeated for each irradiation direction. The modulation of the spot intensity is the aim of plan optimization. The optimal spot intensities result from solving a optimization problem where the user has built a cost function based on dose-volume objectives for the target volume and organs at risk (Figure 2).

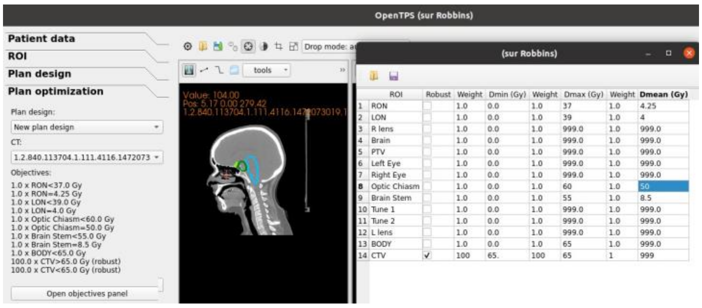

Figure 2. Objectives panel of the OpenTPS GUI.

OpenTPS comes with several built-in optimizers that solve the above problem. It is also designed so that other optimizers may be easily implemented. A variety of iterative solvers are provided: quasi-Newton’s methods from Scipy, in-house implementation of LBFGS algorithms [10], gradient-based methods such as the classical gradient descent and the FISTA algorithm [11] from PyUNLockBox [12]. FISTA solves non-differentiable convex optimization problems by proximal splitting. For example, it can be of particular interest for ArcPT treatment plan optimization where one wants to maximize the spot sparsity by adding a l1-

norm penalty to the total objective function. Linear programming [13] methods are also available and can be used to solve the optimization problem. Based on the list of objectives in the plan design, a model is built with linear equations instead of the typical quadratic version. The model and the solver are built on the Gurobipy package [14], a Python interface to the powerful Gurobi solver, which requires a specific license. Gurobi includes a variety of algorithms to solve linear programming problems.

Treatment evaluation

In OpenTPS, a dose-volume histogram (DVH) evaluation is performed to assess the quality of the treatment plan. For proton therapy, it is important to assess the robustness of the plan to treatment uncertainties, thus robustness evaluation must also be performed on different uncertainty scenarios and not only on the nominal case, leading to a DVH-band instead. Two strategies are available depending on the scenarios definition. The first method, which is based on good practice rules and is commonly used in clinical routine, involves verifying the dose in the error space for a small number of extreme scenarios [15]. In OpenTPS, the dose is re-computed with MCsquare for 81 scenarios resulting from a combination of the setup and proton range erros. The second strategy generates a confidence interval in the so-called dosimetric space. It refers to all possible realizations of the treatment plan, or the different dose distributions that could be achieved. Described in [16], the scenarios are randomly sampled by MCsquare. The analysis in the dosimetric space reports the percentage of evaluation scenarios that meet a certain criterion such as minimum dose coverage for a target percentage (e.g., 95% coverage for 90% of scenarios).

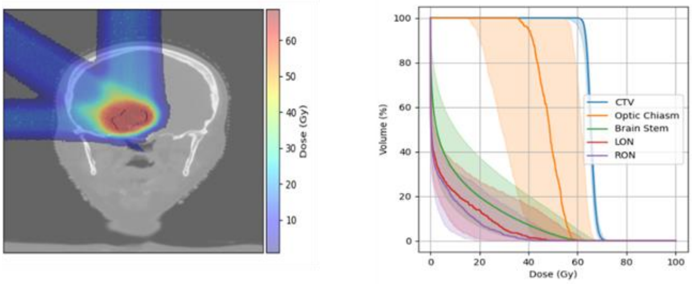

Figure 3. Example of a brain patient optimized in OpenTPS and robust evaluation of the plan (right) with the DVH-bands encompassing all uncertainty scenarios.

3 Results

Use case 1) Optimization in arc proton therapy.

Several solvers were implemented in OpenTPS to optimize arc proton therapy plans. Here, the arc proton problem is formulated as a MIP program and generates the best plan for a brain patient with only five upward energy switching (Figure 4). Details of the implementation and results can be found elsewhere [17].

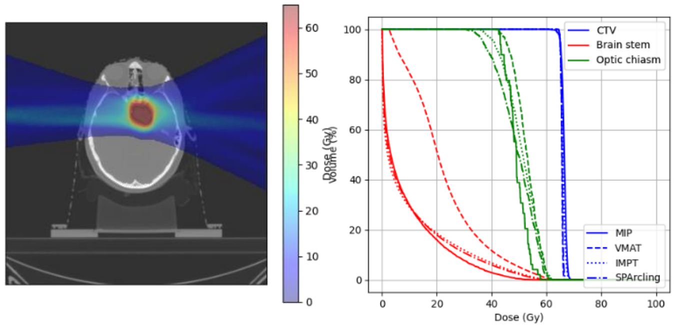

Figure 4. Left: Arc proton therapy dose distribution optimized in OpenTPS with Mixed-Integer Programming (MIP). Right: DVH comparison for Arc proton plans (in-house MIP optimizer, solid line and SPArc [18] technique re-implemented in OpenTPS, dashdot line), VMAT plan (dashed line) and conventional 2 lateral beam IMPT plan (dotted line).

Use case 2) Simulation of different tumor motion scenarios

Using a 4DCT based patient-specific breathing motion model, we can create sequences of variable length containing any desired variation of breathing pattern [19], as well as inter-fraction changes such as setup error simulations or target shrinks and baseline shifts. This way, the delivery of treatment plans can be simulated on these image sequences to assess the plan robustness against the chosen variation. This method of data augmentation is also used to train AI models [20].

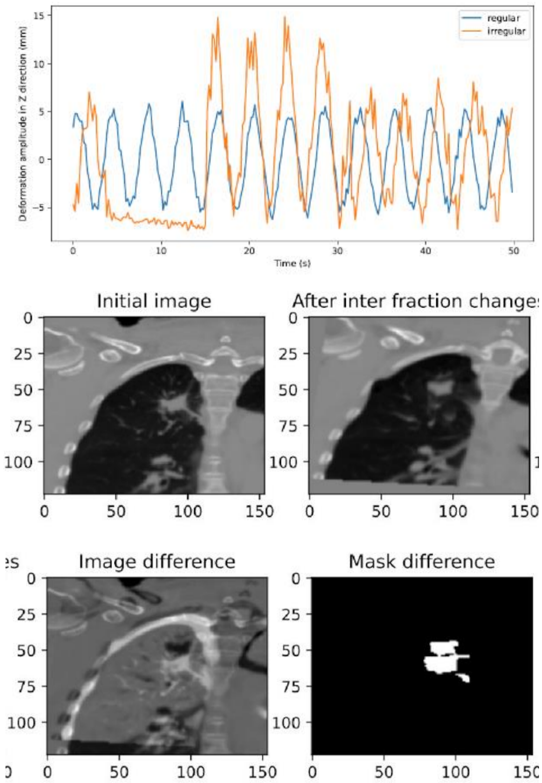

Figure 3. Example of synthetic breathing signals (regular and irregular), generated with OpenTPS tools (top), and example of synthetic inter-fraction deformations applied to and lung CT (bottom).

4 Discussion

OpenTPS is a unique platform for research in radiotherapy treatment planning, and particularly for proton therapy planning, in its current state. It is an open-source tool written in a user-friendly programming language. This allows for both internal contributions and international collaboration to enhance the platform quickly, making it increasingly sophisticated over time. Our research group is currently focusing on several exciting areas within OpenTPS, including arc proton therapy treatment planning and FLASH proton therapy. Depending on the needs of the users, new features might be added in the future such as (but not limited to) other treatment or image modalities, objects tracking or position prediction in dynamic sequences. Additionally, the platform has potential for real-time treatment delivery visualization and 2D to 3D image reconstruction.
The information to get started are available on our website (opentps.org) or on gitlab.

5 Conclusion

OpenTPS offers a flexible and dynamic platform for exploring new possibilities and pushing the boundaries of what is possible in radiation therapy and in proton therapy particularly.

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